

TOPS Technologies – N+ Network Security

Answer Sheet

Theme: Startup Office IT Setup

Section A: Scenario-Based Concept Application

1. CPU, RAM, and storage selection

Choose a CPU that matches the motherboard socket/chipset and can handle multitasking; for development plus video conferencing, a modern multi-core processor with enough threads is preferred. Select RAM in matched kits, verify the motherboard's supported type, speed, and capacity, and leave headroom for multitasking; 16 GB is a practical minimum and 32 GB is better for heavier workloads. For storage, use an SSD, ideally NVMe, for fast boot and application loading, and ensure the motherboard has the correct M.2 or SATA support. Check PSU wattage, cooling, and BIOS support so the parts work together without bottlenecks.

2. Resource allocation and bottlenecks

Background software such as antivirus can consume CPU cycles, RAM, disk I/O, and sometimes network bandwidth. If any one resource becomes saturated, the whole system slows down because processes wait in a queue. A typical example is an antivirus real-time scan using high disk I/O, which makes file access and application launches feel sluggish. The result is not just lower speed, but uneven performance when the workload spikes.

3. Troubleshooting peripheral failures

Start by confirming the device, cable, and port are working on a different workstation. Then check Device Manager or system logs for driver errors, reinstall or update the driver, and compare the OS version and updates on the affected machines. If the issue appears only on one system, inspect USB/power settings, device permissions, and hardware conflicts. If the device fails everywhere, the peripheral itself is likely defective.

4. UPS and surge protection

A UPS keeps the system running long enough to save work and shut down cleanly during power loss, while a surge protector blocks voltage spikes and transient noise. Together they reduce the risk of abrupt shutdowns, file system corruption, and data loss. They also improve hardware reliability by protecting drives, motherboards, and power supplies from unstable power.

Section B: Practical Application

Task 1: Physically assemble a workstation and document the component installation sequence.

Recommended sequence: prepare the case, install CPU, apply thermal paste if needed, mount cooler, insert RAM, fit SSD/HDD, install motherboard in the case, connect PSU cables, add graphics card and other expansion cards, then connect front-panel, USB, audio, and fan headers. After wiring, perform a first boot, enter BIOS/UEFI, verify hardware detection, and then proceed with OS installation.

Task 2: Troubleshoot and resolve a RAM mismatch error within BIOS/UEFI settings.

Power off the system, reseat the modules, and confirm the sticks are installed in the correct slots for dual-channel operation. Enter BIOS/UEFI and check whether the RAM speed, voltage, and timing are set to auto or match the module specifications. Disable an unstable XMP/EXPO profile if needed, update BIOS/UEFI, and test one module at a time to isolate a faulty stick or slot.

Task 3: Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet.

Use one switch, connect five PCs and one printer, and assign addresses from a Class C private network such as 192.168.1.0/24. Example IPs: PC1 192.168.1.10, PC2 192.168.1.11, PC3 192.168.1.12, PC4

192.168.1.13, PC5 192.168.1.14, Printer 192.168.1.20, all with subnet mask 255.255.255.0 and default gateway left blank if no router is used. Verify connectivity using ping between the devices and print a test page if supported.

Task 4: Perform a clean OS installation with manual disk partitioning and startup optimization.

Boot from the installation media, choose custom installation, delete old partitions if required, and create separate partitions for system, data, and swap/page file as appropriate. After installation, install chipset, graphics, network, and storage drivers, apply updates, and disable unnecessary startup applications. Finally, confirm that the OS boots cleanly and that essential devices are recognized.

Section C: Integrated Case Study

Scenario: At Nexus Innovations, three rendering workstations pass morning POST checks but restart randomly during heavy afternoon workloads. The units are stored in poorly ventilated desk cabinets.

1. Three likely hardware components

The most likely components are the CPU, the PSU, and the RAM. Under load, overheating CPU cores, unstable power delivery, or marginal memory modules can all trigger resets.

2. Why POST passes in the morning

POST only checks basic hardware initialization at startup, when temperatures and load are low. In the afternoon, accumulated heat and sustained workload push components beyond safe limits, causing thermal shutdowns or power instability.

3. Using a stress-test tool

Run a CPU, GPU, and memory stress test while monitoring temperatures, voltages, and fan speeds with hardware monitoring software. If the system reboots or throttles only under stress, that strongly suggests a thermal or power delivery problem. Record the exact temperature and time at failure for comparison.

4. Three remediation steps

Improve cabinet ventilation and clean dust from heatsinks and filters. Replace old thermal paste and verify all fans are spinning at proper speed. If needed, redistribute workload, improve cable management for airflow, and upgrade cooling or PSU based on test results.

5. PSU upgrade justification

A PSU upgrade is necessary if stress testing shows voltage drops, shutdowns under peak load, or the current unit is below the required wattage for the installed hardware. It may not be necessary if the problem is clearly thermal and the PSU remains stable during load tests. In that case, better cooling and airflow would be the first fix.

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